Robert Chen

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Education

University of Virginia | GPA: 3.6/4.0 | Charlottesville, VA
B.S. in Computer Engineering and Electrical Engineering (Double Major) | May 2026

Experience

Software Engineer (Analyst) — Arlington, VA
Accenture Federal Services — *Aug 2026 – Present*

- Designed the architecture for a benefits processing microservice, analyzing about **1,800 business rules** to determine when rules could run once or needed reevaluation.
- Implemented a Java/Spring Boot REST service that invokes a rules engine to evaluate rules from an AI pipeline. Matched legacy results in **all 28 API integration tests** for a benefits workflow and added Excel-based regression tests to the Maven build.
- Developed a debugger to trace rule execution through the API and investigate unexpected benefits decisions.
- Improved notification error reporting and standardized phone formats across 3 Spring Boot services, adding unit tests.

DevOps Cloud Automation Engineer Intern — Fairfax, VA
Verint — *May 2025 – Nov 2025*

- Deployed a secure AWS Transfer Family web client on ECS Fargate using CloudFormation, Cognito, and Angular.
- Won **Best Intern Project** out of 6 teams for a banking assistant prototype with voice biometrics and compliance guardrails. Identified feature gaps for a business case projecting about $10M in annual savings.
- Cut CI/CD checks from 3–5 to **2–3 minutes** for about 30 pull requests per week using GitHub Actions and Harness.
- Improved PR-check precision by **12%** by curating pull-request data and tuning an Azure AI Foundry code review model.
- Migrated Terraform workspaces to OpenTofu, standardizing remote state and reusable modules for EC2, S3, and IAM.

Research Assistant (ML & Computer Vision) — Charlottesville, VA
Visual Intelligence Lab, University of Virginia — *Sep 2025 – May 2026*

- Parsed tennis broadcasts into JSON match data with Python, OpenCV, Tesseract OCR, smoothing, and finite-state logic.

Software Engineering Intern — Centreville, VA
S&J Chen, LLC — *May 2024 – Aug 2024*

- Reduced manual rent follow-ups by about **60%** across 2 shopping centers by shipping a full-stack tenant payment portal with Next.js, TypeScript, Supabase, and Vercel.
- Cut monthly reporting from about 3 hours to **30 minutes** by automating Stripe webhook reconciliation and CSV exports.

Projects

Badminton Swing Analyzer | *Embedded C, Flutter, BLE, IMU, SQLite*

- Built Embedded C firmware and a Flutter app using 100 Hz IMU and audio data, achieving about **95%** hit-vs-shadow-swing accuracy in on-court tests with FFT and physics-based features.

Photo Curator App | *React Native, TypeScript, TensorFlow.js, NestJS, AWS*

- Built a photo curator with swipe review, blur detection, facial recognition, embedding-based grouping, secure sync, and **sub-300 ms** similarity search (NestJS, JWT, pgvector, S3).

Outreach Ops Agent | *n8n, OpenAI API, Google Sheets, Clay, Gmail API, Docker*

- Built an n8n outreach workflow with GPT-4.1 lead research, structured JSON output, LLM draft scoring, and human approval before sending, cutting hands-on prep from **about 15 minutes to 1 minute** per posting in a timed trial.

Technical Skills

Programming Languages: Java, Python, TypeScript, JavaScript, SQL, C, C++
Frameworks & Tools: Spring Boot, REST APIs, OpenAPI, Maven, JUnit, Mockito, Kafka, React, Next.js, Angular, Node.js, Git
ML & Data: PostgreSQL, pgvector, Supabase, OpenCV, Tesseract OCR, TensorFlow.js, Azure AI Foundry, NumPy, Pandas
Cloud & DevOps: AWS (ECS, S3, Cognito), Docker, Kubernetes, GitHub Actions, Harness, Terraform, OpenTofu, Linux

Leadership & Activities

Leadership: Yo-Yo Club, Founder & President (about 50 members); Taiwanese Student Association (TSA), Sports Chair
Athletics: Club Pickleball (Collegiate Nationals); Club Tennis (Competitive Travel Team); Club Figure Skating (2× 1st Place)